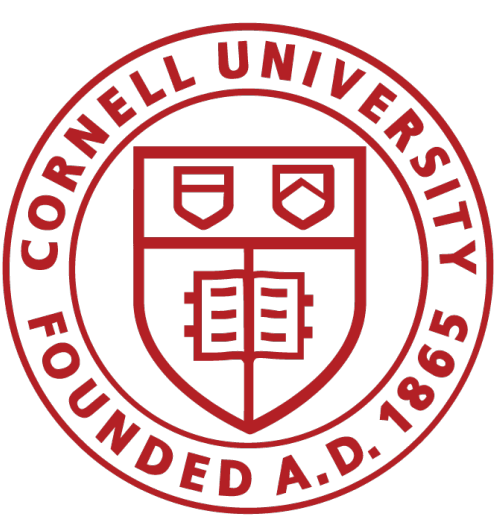


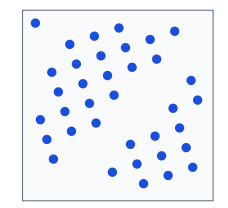
# From Mimetic to Structured Initialization: Reproducing Attention Biases for Small-Data Vision Transformers

Lu Shen, Xuanbo Jia, Yichen Li

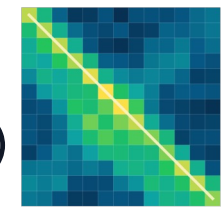
Course Project | Vision Transformers | Small-Data Learning



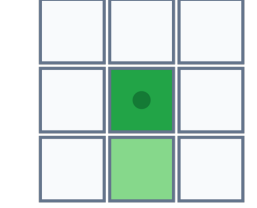
Default / Random Initialization



Mimetic Initialization  
(Mimic Pretrained Attention Patterns)



Structured Impulse Initialization  
(Program CNN-like Local Attention Bias)



**Goal:** Inject useful inductive bias via initialization to help ViTs learn from limited data.

## 1. MOTIVATION

- Vision Transformers work well at scale but are difficult to train from scratch on small datasets.
- CNNs generalize better in low-data regimes because they encode locality and translation bias.
- Question: can we inject useful inductive bias through initialization alone, without changing the ViT architecture?

## 2. RESEARCH QUESTION

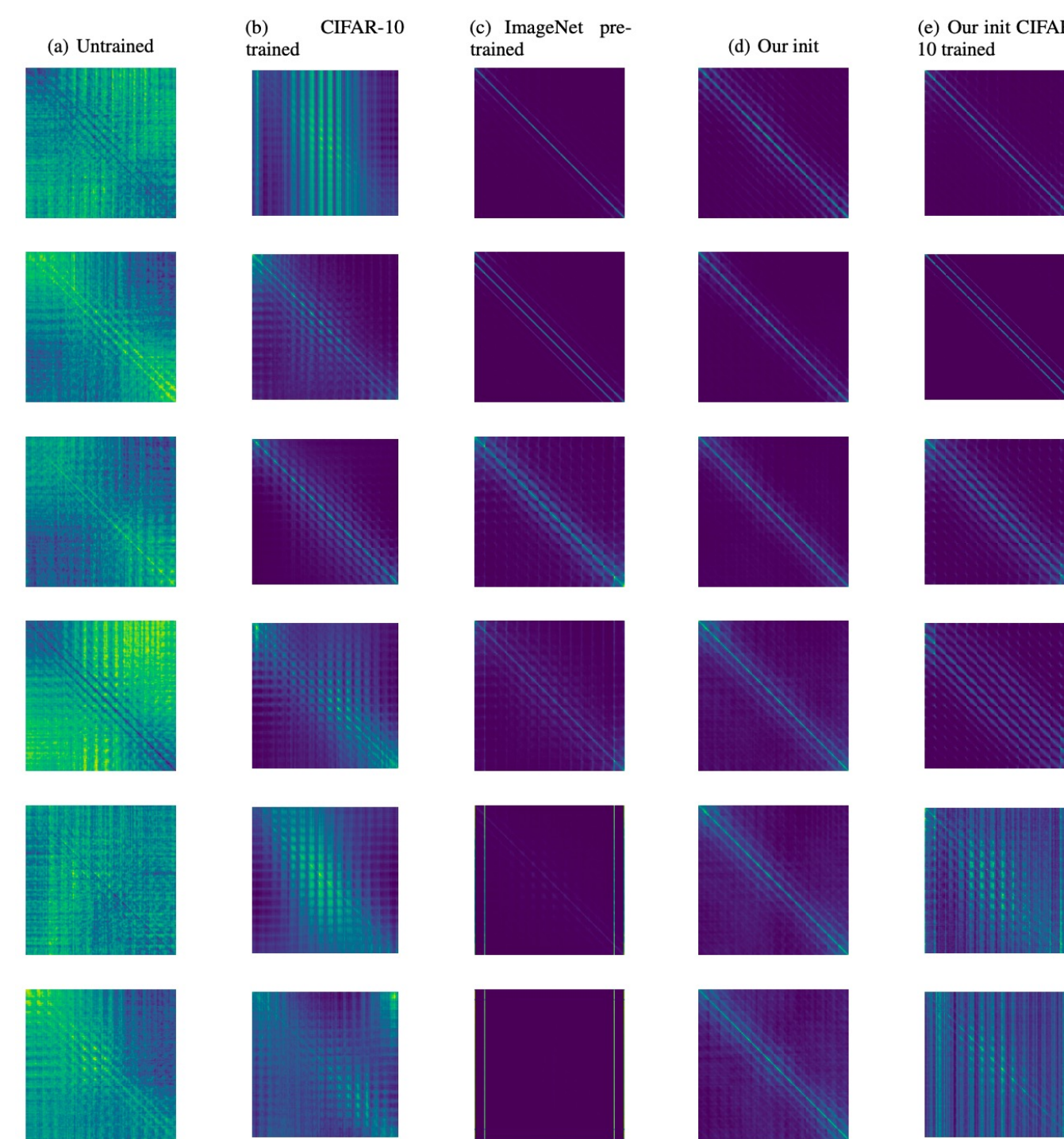
- Can attention bias be introduced purely through initialization?
- Compare default ViT initialization, Mimetic Initialization, and Structured Initialization.
- Focus on small-data image classification settings.
- Chosen result: reproduce the CIFAR-style comparison among default, Mimetic, and Structured impulse initialization, like Figure-style attention map visualization.

## 5. OUR RE-IMPLEMENTATION

|  |                |                                                                   |
|--|----------------|-------------------------------------------------------------------|
|  | <b>Dataset</b> | CIFAR-10, CIFAR-100; 10% / 25% subsets                            |
|  | <b>Model</b>   | ViT-Tiny (from timm implementation)                               |
|  | <b>Method</b>  | Default / Structured / Mimetic, Structured Impulse Initialization |
|  | <b>Metrics</b> | Top-1 accuracy, training curves, attention-map visualizations     |
|  | <b>Tools</b>   | PyTorch, timm, Colab A100                                         |

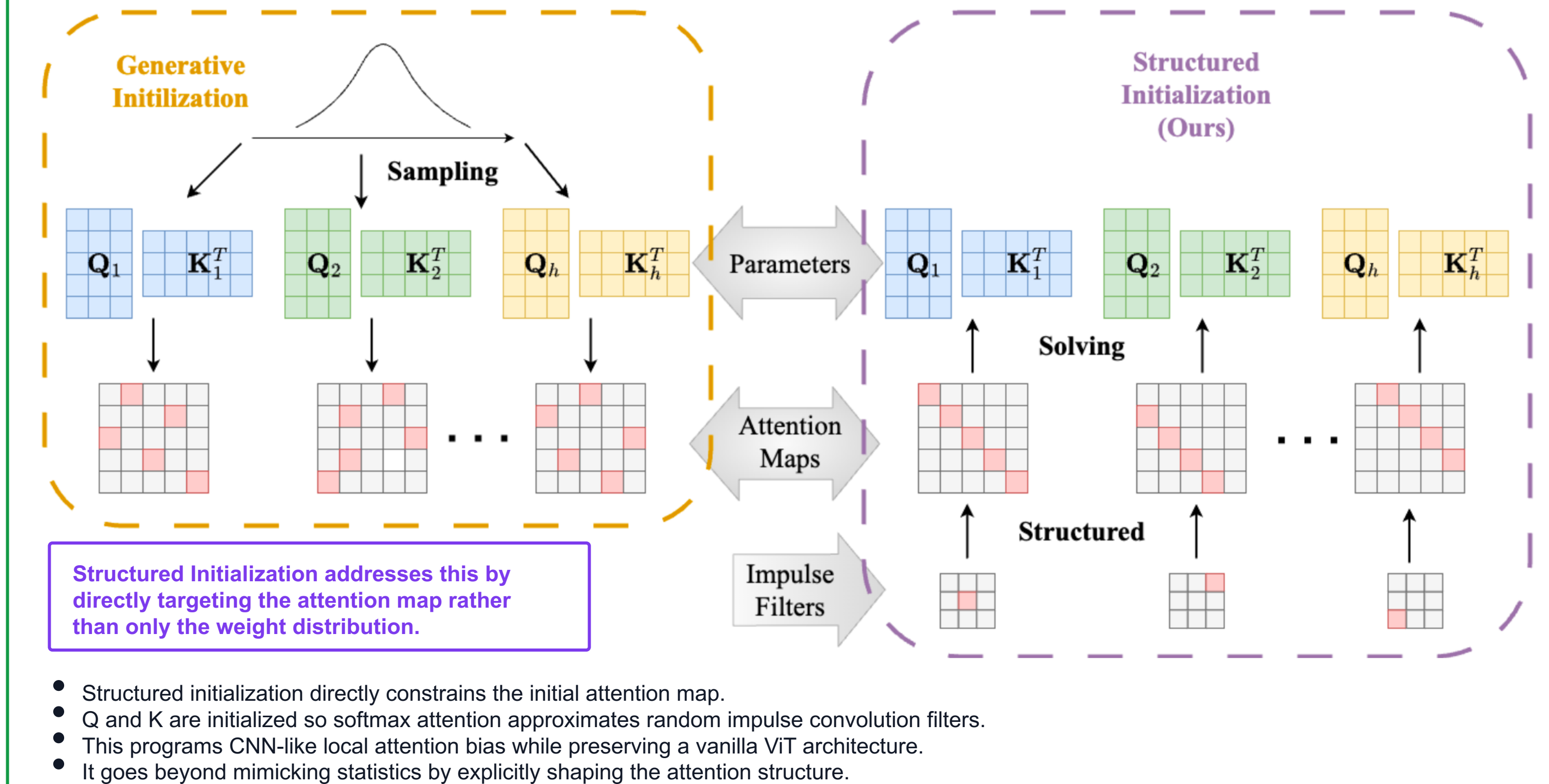
## 3. PRIOR WORK: MIMETIC INITIALIZATION

Mimetic Initialization of Self-Attention Layers



- Mimetic initialization studies pretrained ViTs.
- Pretrained attention layers show repeatable weight-product statistics.
- These statistics lead to diagonal / banded attention-map patterns.
- Mimetic initialization copies pretrained-like statistics into new attention layers.
- Strength: simple, data-driven, no architecture change.
- Limitation: mimics pretrained statistics, but does not directly specify the desired spatial attention structure.

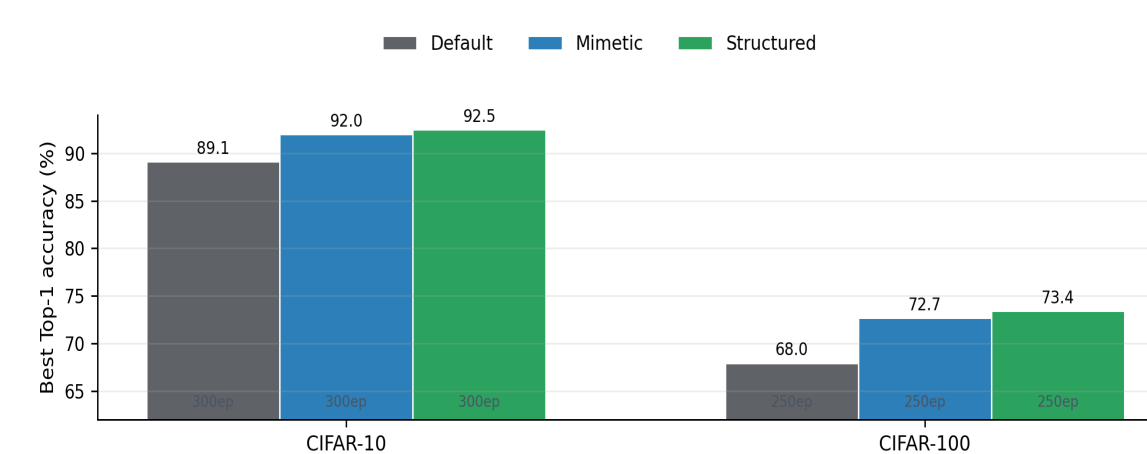
## 4. FURTHER STEP: STRUCTURED INITIALIZATION



## 6. OUR REPRODUCTION RESULTS

### 6.1 Full-Data Reproduction

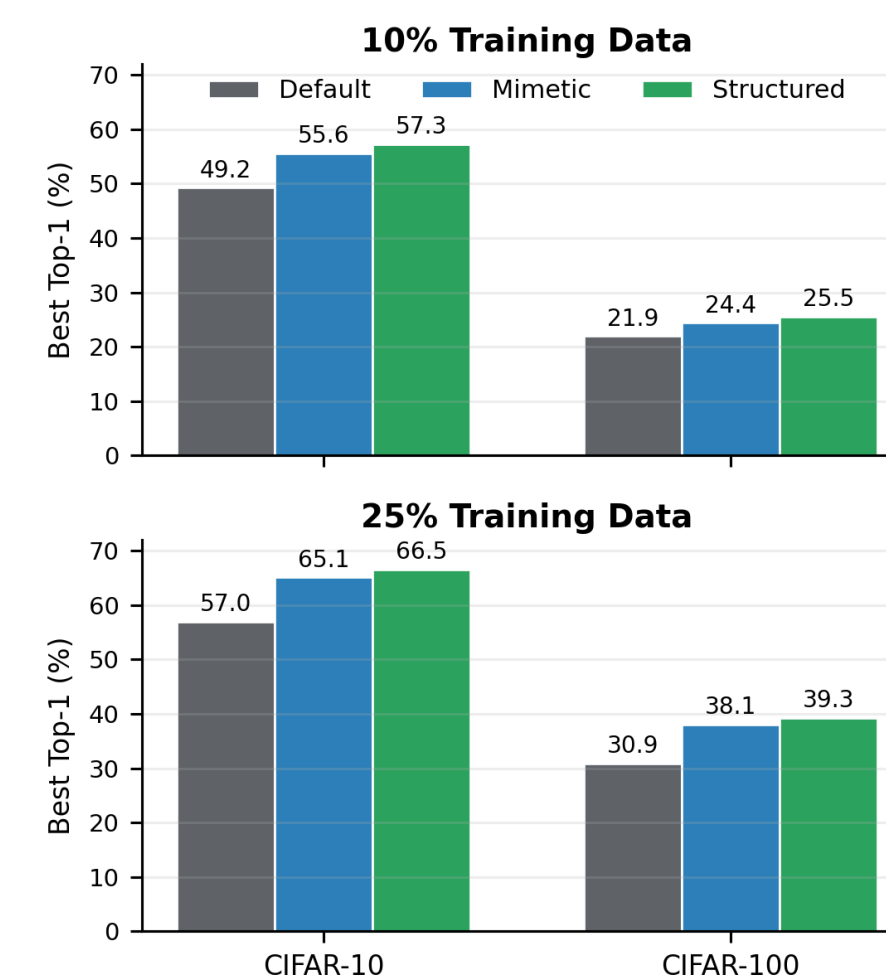
Full-Data Reproduction: Accuracy and Seed Stability



| CIFAR-10 Seed Stability at 250 Epochs |        |        |        |                |  |
|---------------------------------------|--------|--------|--------|----------------|--|
| CIFAR-10 Init                         | Seed 0 | Seed 1 | Seed 2 | Mean +/- Std   |  |
| Default                               | 88.20  | 87.72  | 87.44  | 87.79 +/- 0.38 |  |
| Mimetic                               | 91.32  | 91.03  | 90.42  | 90.92 +/- 0.46 |  |
| Structured                            | 91.85  | 90.90  | 89.92  | 90.89 +/- 0.97 |  |

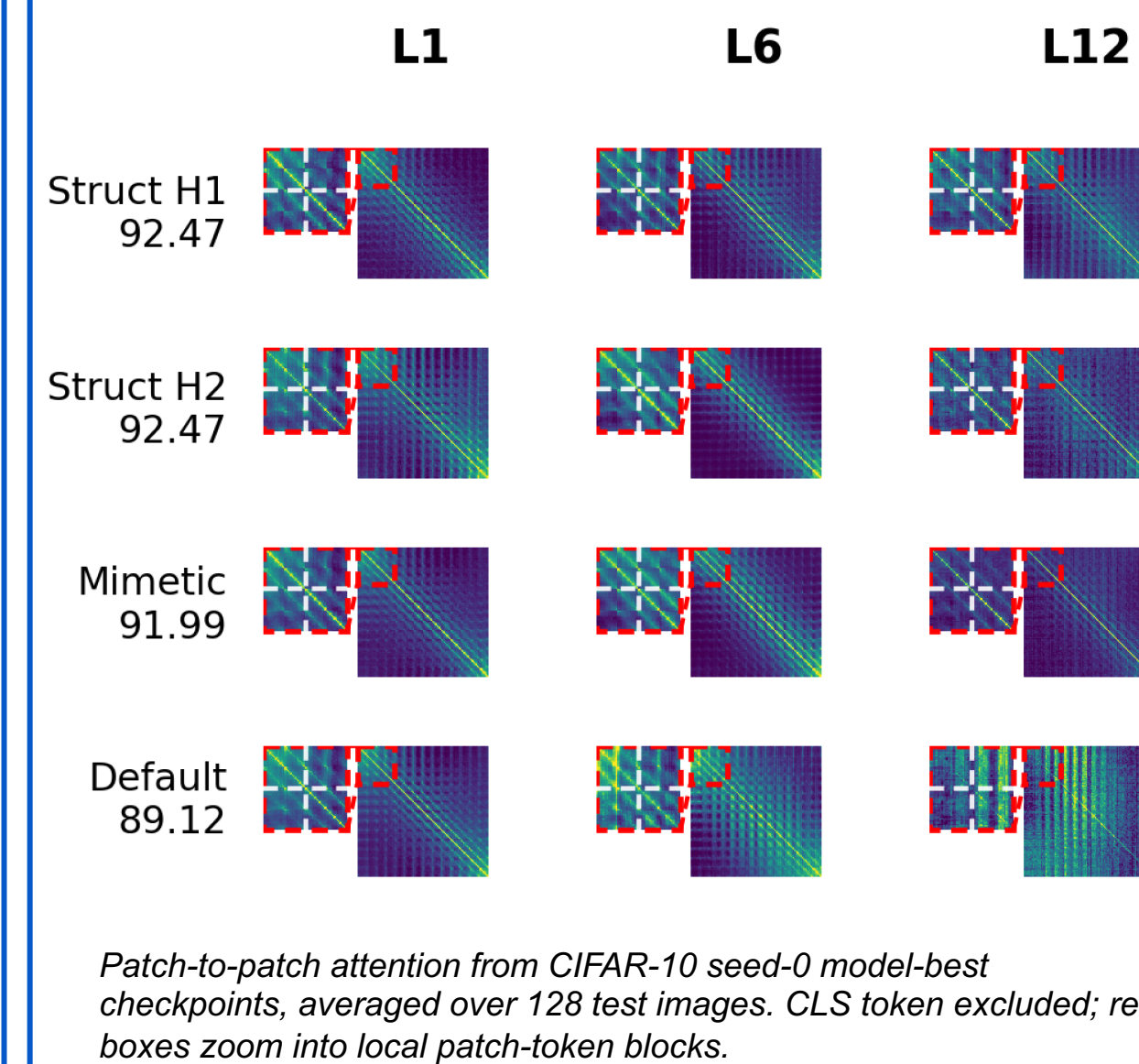
Common 250-epoch budget. CIFAR-10 reports mean +/- std over seeds 0, 1, 2; CIFAR-100 uses seed 0

### 6.2 Low-Data Validation



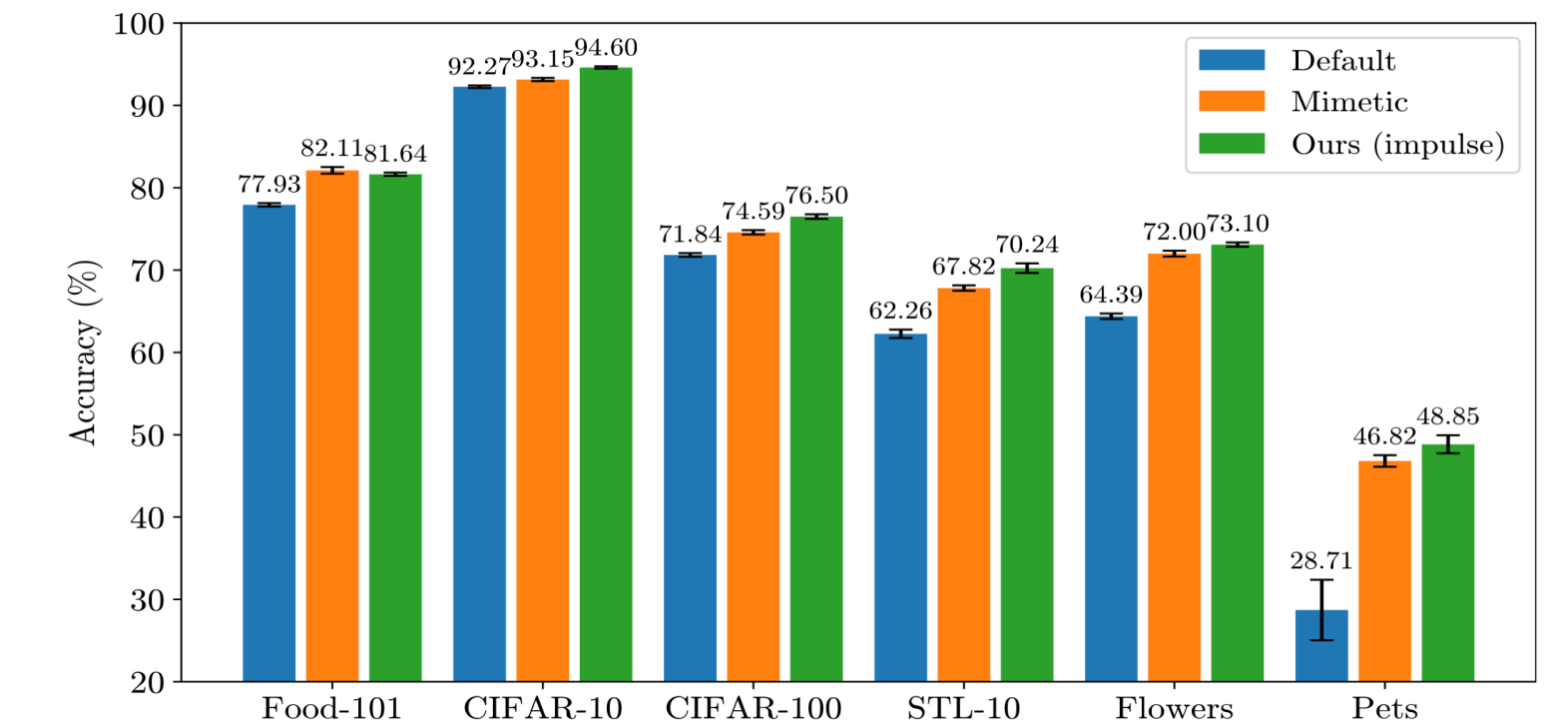
Seed 1, 150 epochs, batch size 512. Validation uses the full test split.

### 6.3 Trained Attention Maps

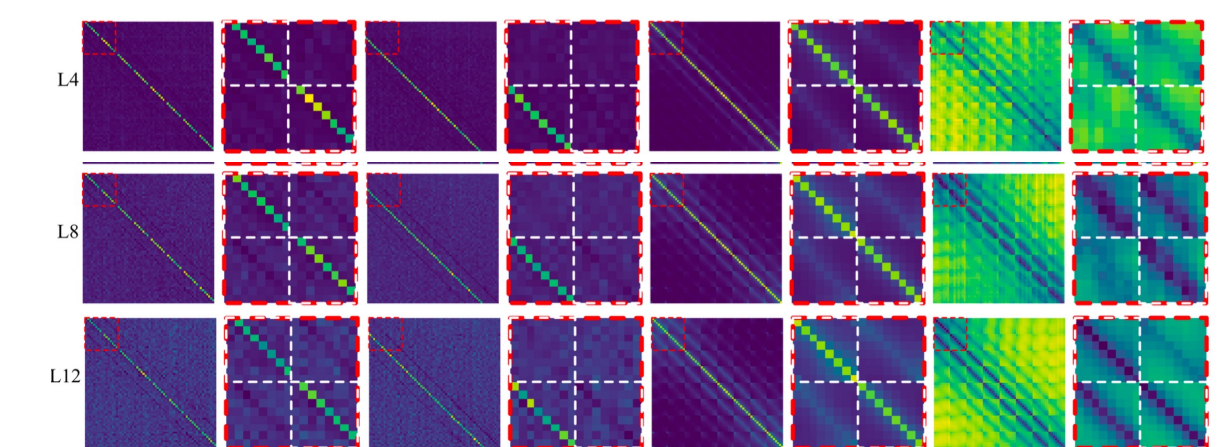


## 7. SUPPORTING EVIDENCE FROM PRIOR WORK

### 7.1 Reported Performance Trends (Prior)



### 7.2 Example Attention Structures (Prior)



## DISCUSSION

Mimetic and Structured improve over Default; low-data experiments show larger gains under reduced data.

## TAKEAWAYS

Initialization injects useful inductive bias; structured attention changes attention patterns; results are sensitive to recipe and seed.

## REFERENCES

- Trockman & Kolter, Mimetic Initialization of Self-Attention Layers, ICML 2023.
- Zheng et al., Structured Initialization for Vision Transformers, NeurIPS 2025.
- Wightman, PyTorch Image Models / timm, 2019.